



EdgeFire – How to Set Up Site-to-Site VPN between TXOne EdgeFire and Palo Alto Devices

Application Note

August 2023

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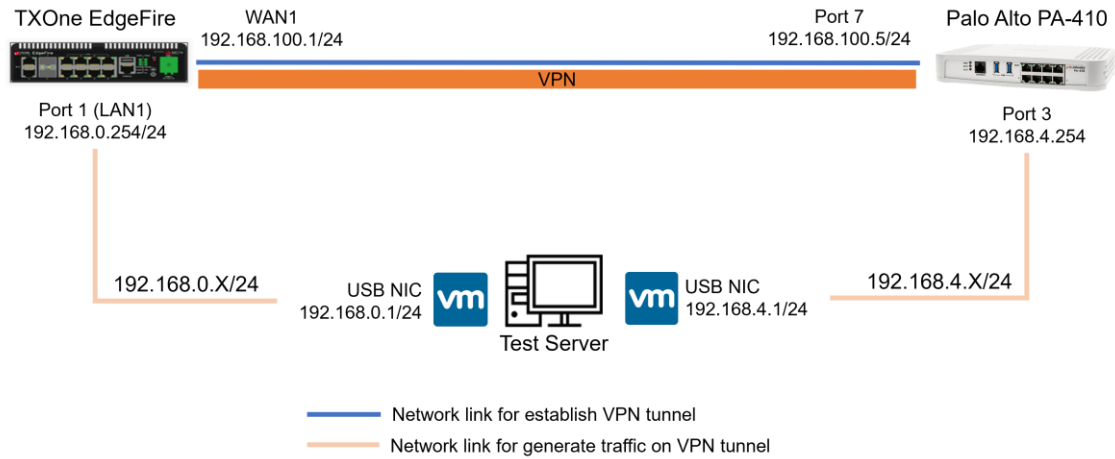
- EdgeIPS
- EdgeIPS Pro
- EdgeIPS LE
- EdgeFire
- EdgeOne
- OT Defense Console

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1 Network Topology



Note: The management interface (WAN2 for EdgeFire and MGT for PA-410) and the console connection are not shown on the figure

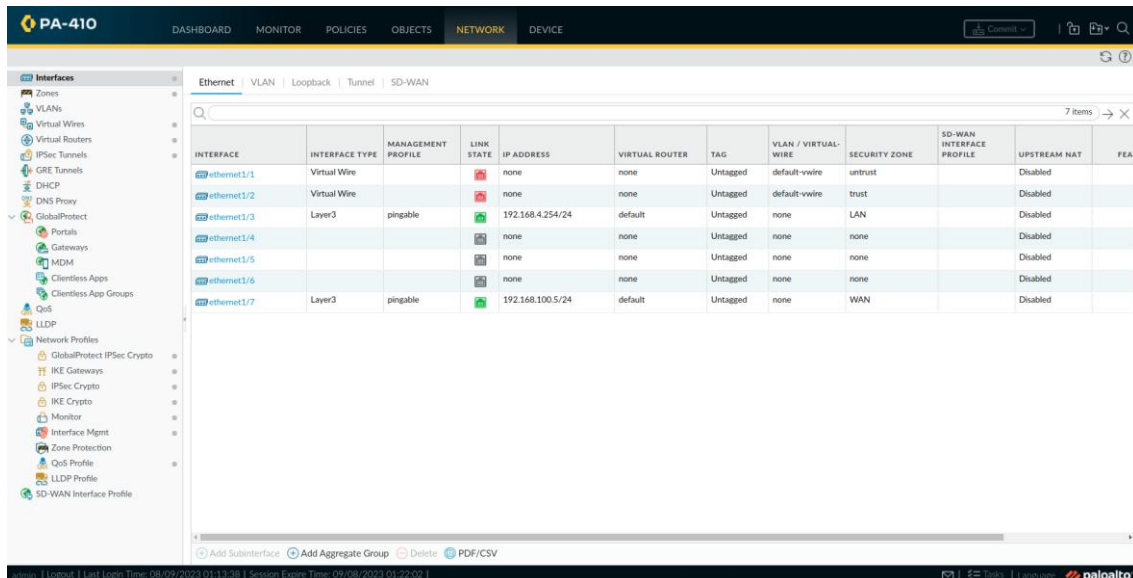
2 Setup flow for Palo Alto PA-410

2.1 Setting network interfaces

2.1.1 CLI Commands

```
set network interface ethernet ethernet1/3 layer3 ndp-proxy enabled no
set network interface ethernet ethernet1/3 layer3 sdwan-link-settings
upstream-nat enable no
set network interface ethernet ethernet1/3 layer3 sdwan-link-settings
upstream-nat static-ip
set network interface ethernet ethernet1/3 layer3 sdwan-link-settings
enable no
set network interface ethernet ethernet1/3 layer3 ip 192.168.4.254/24
set network interface ethernet ethernet1/3 layer3 lldp enable no
set network interface ethernet ethernet1/7 layer3 ndp-proxy enabled no
set network interface ethernet ethernet1/7 layer3 sdwan-link-settings
upstream-nat enable no
set network interface ethernet ethernet1/7 layer3 sdwan-link-settings
upstream-nat static-ip
set network interface ethernet ethernet1/7 layer3 sdwan-link-settings
enable no
set network interface ethernet ethernet1/7 layer3 ip 192.168.100.5/24
set network interface ethernet ethernet1/7 layer3 lldp enable no
set network virtual-router default interface [ ethernet1/3 ethernet1/7 ]
set zone WAN network layer3 ethernet1/7
set zone LAN network layer3 ethernet1/3
set network profiles interface-management-profile pingable ping yes
set import network interface [ ethernet1/3 ethernet1/7 ]
set network interface ethernet ethernet1/7 layer3 interface-management-
profile pingable
set network interface ethernet ethernet1/3 layer3 interface-management-
profile pingable
```

2.1.2 Web GUI configuration screenshot



The screenshot shows the Palo Alto PA-410 Web GUI. The left sidebar contains a tree view with categories like Interfaces, Zones, Virtual Wires, Virtual Routers, IPSec Tunnels, GRE Tunnels, DHCP, DNS Proxy, GlobalProtect, Portals, Gateways, MDM, Clientless Apps, Clientless App Groups, QoS, LLDP, Network Profiles, GlobalProtect IPSec Crypto, IKE Gateways, IPSec Crypto, IKE Crypto, Monitor, Interface Mgmt, Zone Protection, QoS Profile, LLDP Profile, and SD-WAN Interface Profile. The main content area is titled 'Ethernet' and shows a table of 7 items. The table has columns: INTERFACE, INTERFACE TYPE, MANAGEMENT PROFILE, LINK STATE, IP ADDRESS, VIRTUAL ROUTER, TAG, VLAN / VIRTUAL-WIRE, SECURITY ZONE, SD-WAN INTERFACE PROFILE, UPSTREAM NAT, and FEA. The data rows are as follows:

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE	SD-WAN INTERFACE PROFILE	UPSTREAM NAT	FEA
ethernet1/1	Virtual Wire		none	none	none	Untagged	default-vwire	untrust		Disabled	
ethernet1/2	Virtual Wire		none	none	none	Untagged	default-vwire	trust		Disabled	
ethernet1/3	Layer3	pingable	192.168.4.254/24	default	Untagged	none	LAN			Disabled	
ethernet1/4			none	none	Untagged	none	none			Disabled	
ethernet1/5			none	none	Untagged	none	none			Disabled	
ethernet1/6			none	none	Untagged	none	none			Disabled	
ethernet1/7	Layer3	pingable	192.168.100.5/24	default	Untagged	none	WAN			Disabled	

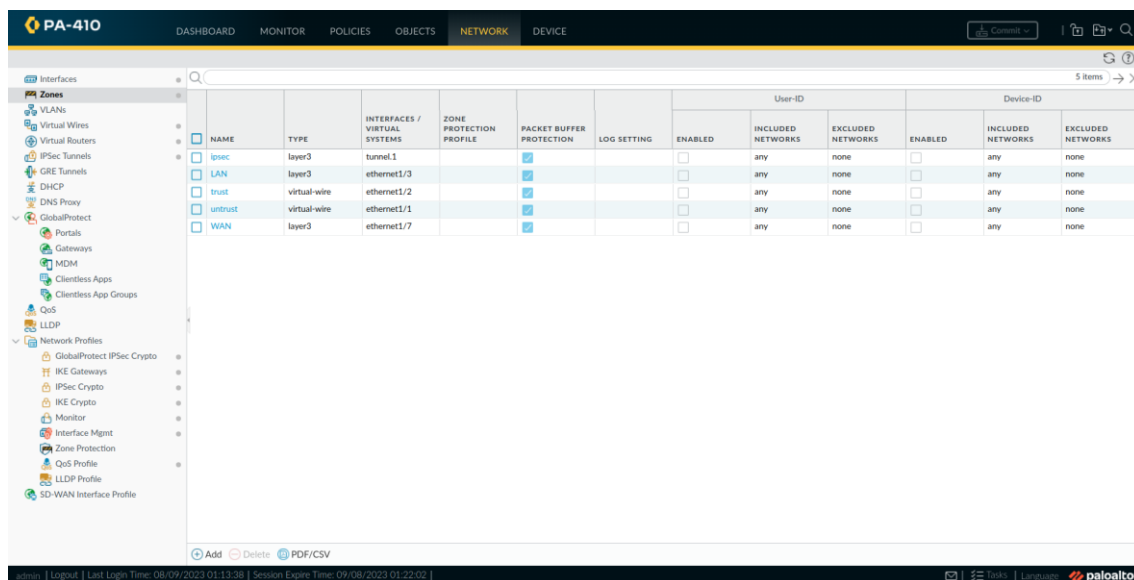
At the bottom of the table, there are links: Add Subinterface, Add Aggregate Group, Delete, and PDF/CSV.

2.2 Setting the tunnel interface and security rules

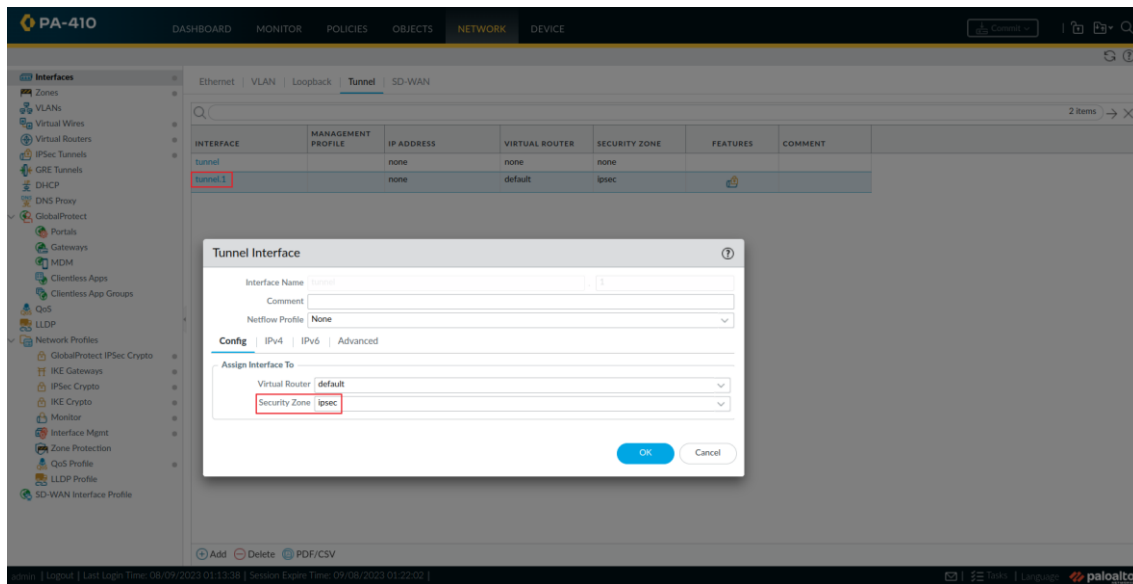
2.2.1 CLI Commands

```
set network interface tunnel units tunnel.1
set zone ipsec network layer3 tunnel.1
set rulebase security rules Taffic-out to ipsec
set rulebase security rules Taffic-out from LAN
set rulebase security rules Taffic-out source any
set rulebase security rules Taffic-out destination any
set rulebase security rules Taffic-out source-user any
set rulebase security rules Taffic-out category any
set rulebase security rules Taffic-out application any
set rulebase security rules Taffic-out service any
set rulebase security rules Taffic-out source-hip any
set rulebase security rules Taffic-out destination-hip any
set rulebase security rules Taffic-out action allow
set rulebase security rules Taffic-out log-start yes
set rulebase security rules Taffic-out log-end yes
set rulebase security rules Traffic-in to LAN
set rulebase security rules Traffic-in from ipsec
set rulebase security rules Traffic-in source any
set rulebase security rules Traffic-in destination any
set rulebase security rules Traffic-in source-user any
set rulebase security rules Traffic-in category any
set rulebase security rules Traffic-in application any
set rulebase security rules Traffic-in service any
set rulebase security rules Traffic-in source-hip any
set rulebase security rules Traffic-in destination-hip any
set rulebase security rules Traffic-in action allow
set rulebase security rules Traffic-in log-start no
set rulebase security rules Traffic-in log-end yes
set import network interface [ ethernet1/3 ethernet1/7 tunnel.1 ]
```

2.2.2 Web GUI configuration screenshot



							User-ID			Device-ID		
NAME	TYPE	INTERFACES / VIRTUAL SYSTEMS	ZONE PROTECTION PROFILE	PACKET BUFFER PROTECTION	LOG SETTING	ENABLED	INCLUDED NETWORKS	EXCLUDED NETWORKS	ENABLED	INCLUDED NETWORKS	EXCLUDED NETWORKS	
ipsec	layer3	tunnel.1		☒		☐	any	none	☐	any	none	
LAN	layer3	ethernet1/3		☒		☐	any	none	☐	any	none	
trust	virtual-wire	ethernet1/2		☒		☐	any	none	☐	any	none	
untrust	virtual-wire	ethernet1/1		☒		☐	any	none	☐	any	none	
WAN	layer3	ethernet1/7		☒		☐	any	none	☐	any	none	



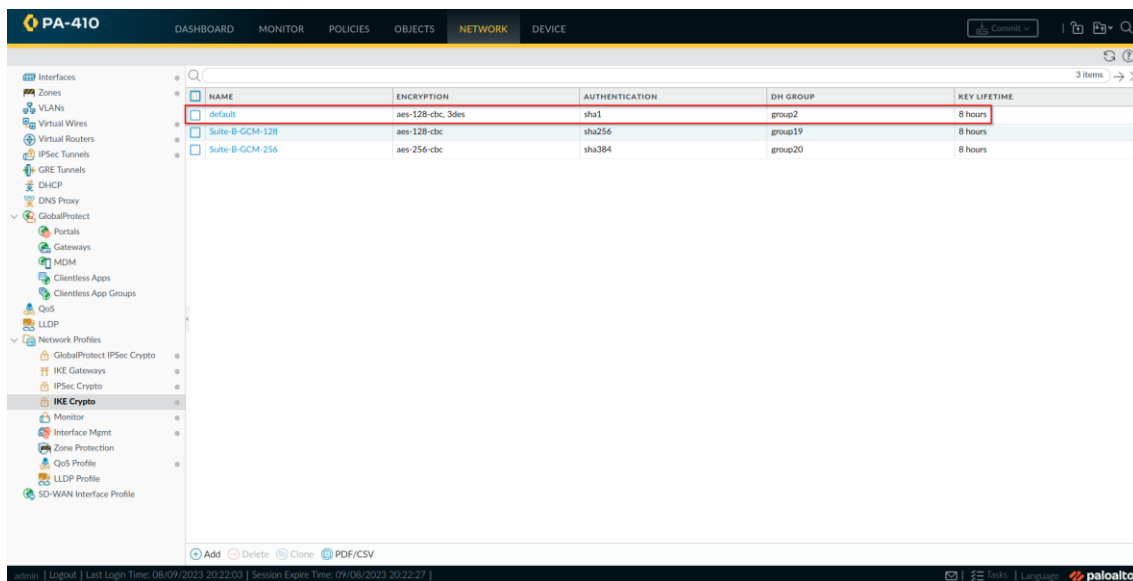
2.3 Setting IKE Crypto (IKE Phase 1)

Use the default profile which is auto created in Palo Alto system

Encryption: aes-128-cbc, 3des

Authentication: sha1

DH group: group 2



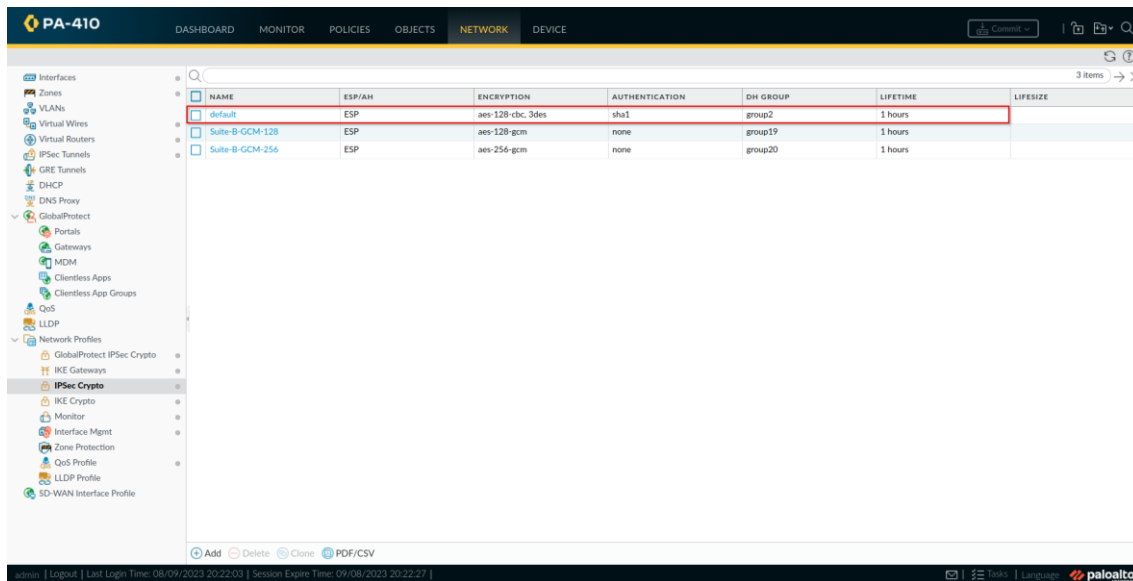
2.4 Setting IPSec Crypto (IKE Phase 2)

Use the default profile which is auto created in Palo Alto system

Encryption: aes-128-cbc, 3des

Authentication: sha1

DH group: group 2



2.5 Setting the IKE Gateway with default IKE Crypto Profile (Step 2.5.1 or Step 2.5.2)

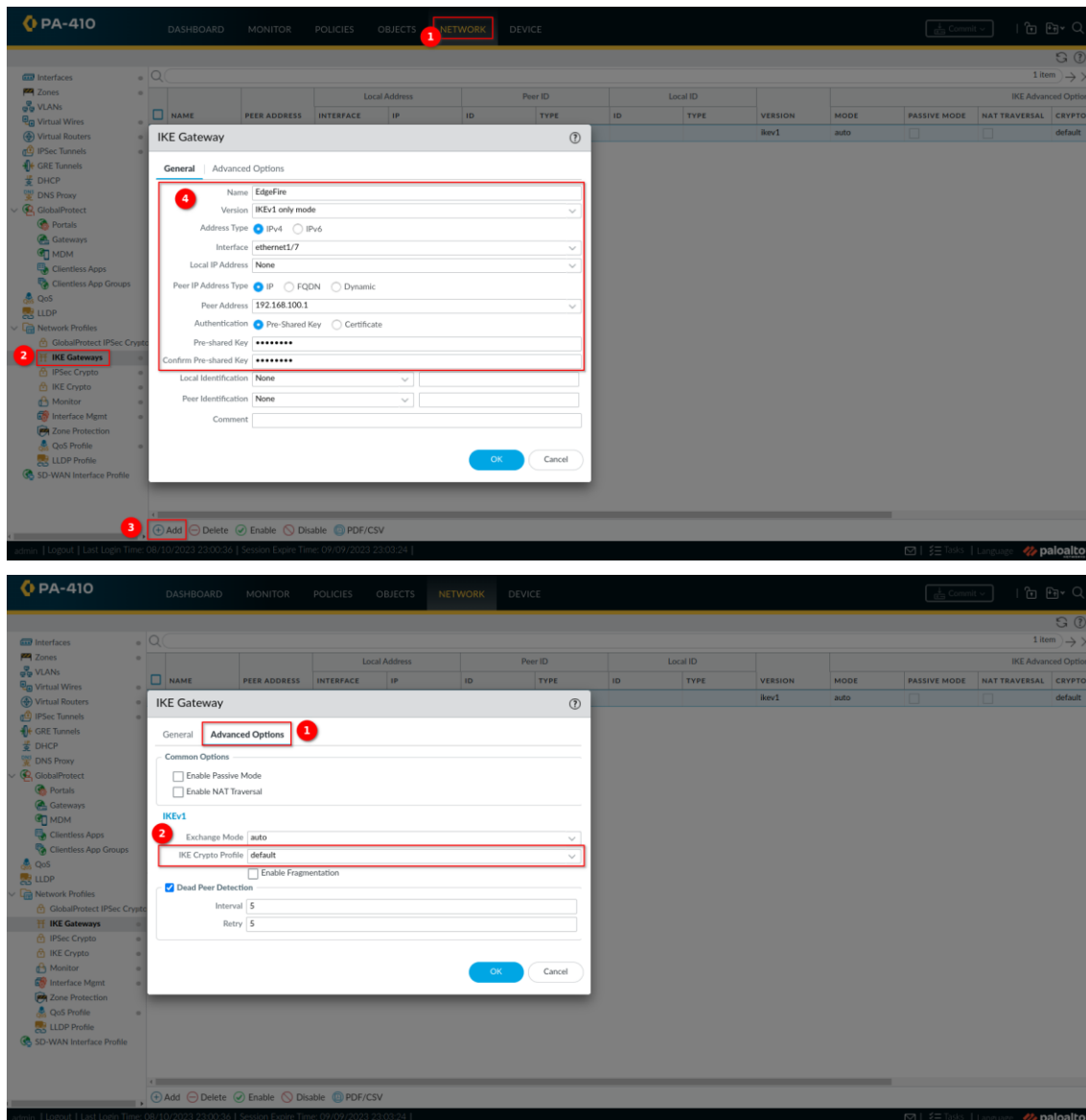
2.5.1 IKE v1

2.5.1.1 CLI Commands

```
set network ike gateway EdgeFire authentication pre-shared-key key
1qaz2WSX
set network ike gateway EdgeFire protocol ikev1 dpd enable yes
set network ike gateway EdgeFire protocol ikev2 dpd enable yes
set network ike gateway EdgeFire protocol version ikev1
set network ike gateway EdgeFire local-address ip 192.168.100.5/24
set network ike gateway EdgeFire local-address interface ethernet1/7
set network ike gateway EdgeFire protocol-common nat-traversal enable no
set network ike gateway EdgeFire protocol-common fragmentation enable no
set network ike gateway EdgeFire peer-address ip 192.168.100.1
```

2.5.1.2 Web GUI configuration screenshot

Web GUI path: Network -> Network Profiles -> IKE Gateways -> Add



The IKE crypto profile setting is mapped to the default profile that the system automatically created (Step 3)

2.5.2 IKE v2

2.5.2.1 CLI Commands

```
set network ike gateway EdgeFire authentication pre-shared-key key
1qaz2WSX
set network ike gateway EdgeFire protocol ikev1 dpd enable yes
set network ike gateway EdgeFire protocol ikev2 dpd enable yes
set network ike gateway EdgeFire protocol version ikev2
set network ike gateway EdgeFire local-address ip 192.168.100.5/24
set network ike gateway EdgeFire local-address interface ethernet1/7
set network ike gateway EdgeFire protocol-common nat-traversal enable no
set network ike gateway EdgeFire protocol-common fragmentation enable no
set network ike gateway EdgeFire peer-address ip 192.168.100.1
```

2.5.2.2 Web GUI configuration screenshot

Web GUI path: Network -> Network Profiles -> IKE Gateways -> Add

The first screenshot shows the 'IKE Gateway' configuration window in the Palo Alto PA-410 interface. The 'General' tab is active. The 'Name' field is set to 'EdgeFire'. The 'Version' dropdown is set to 'IKEv2 preferred mode'. The 'Address Type' is set to 'IPv4'. The 'Interface' is set to 'ethernet1/7'. The 'Local IP Address' is set to 'None'. The 'Peer IP Address Type' is set to 'IP'. The 'Peer Address' is set to '192.168.100.1'. The 'Authentication' is set to 'Pre-Shared Key'. The 'Pre-shared Key' and 'Confirm Pre-shared Key' fields are filled with asterisks. The 'Local Identification' and 'Peer Identification' are both set to 'None'. The 'Comment' field is empty. The 'OK' button is highlighted.

The second screenshot shows the 'IKE Gateway' configuration window in the Palo Alto PA-410 interface. The 'Advanced Options' tab is active. The 'Common Options' section has 'Enable Passive Mode' and 'Enable NAT Traversal' unchecked. The 'IKEv1' section has 'Exchange Mode' set to 'auto'. The 'IKE Crypto Profile' dropdown is set to 'default'. The 'Dead Peer Detection' checkbox is checked, with 'Interval' and 'Retry' both set to '5'. The 'Enable Fragmentation' checkbox is unchecked. The 'OK' button is highlighted.

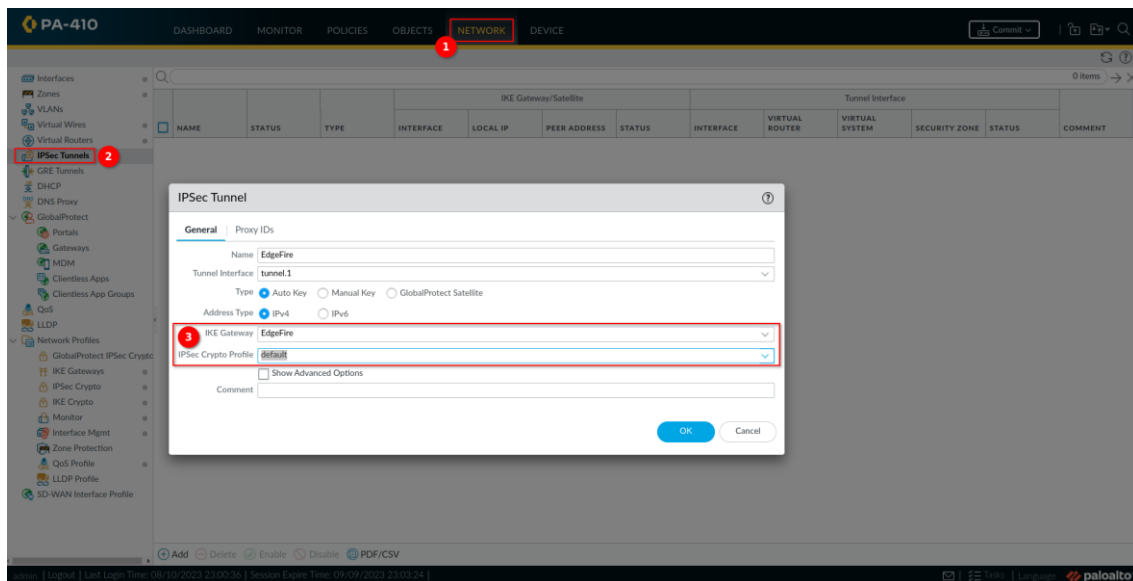
The IKE crypto profile setting is mapped to the default profile that the system automatically created (Step 3)

2.6 Setting the IPsec tunnel

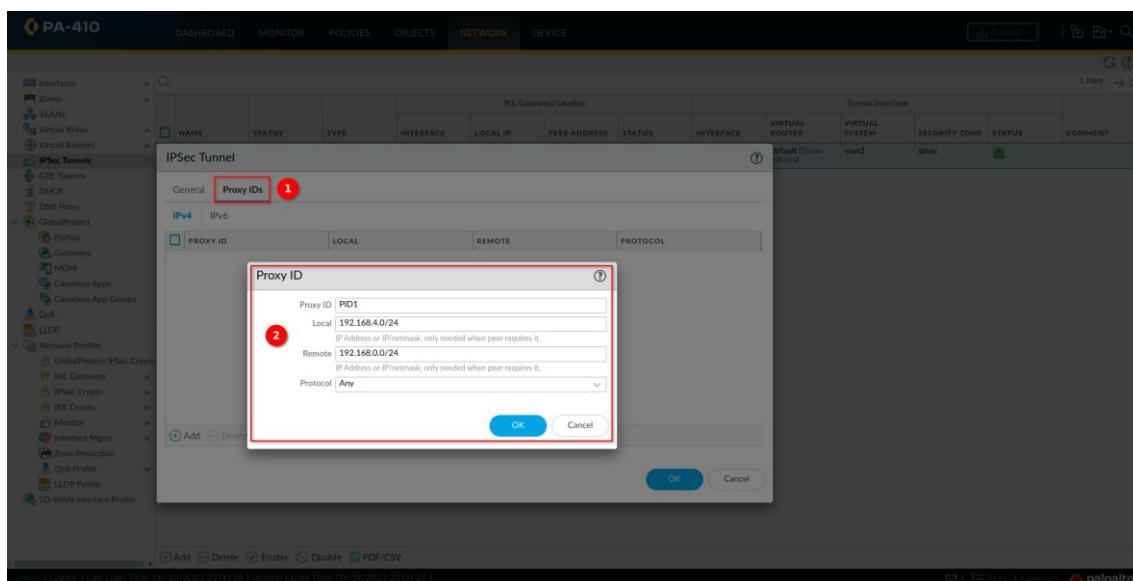
2.6.1 CLI Commands

```
set network tunnel ipsec IPSEC1 auto-key ike-gateway EdgeFire
set network tunnel ipsec IPSEC1 auto-key proxy-id ID1 protocol any
set network tunnel ipsec IPSEC1 auto-key proxy-id ID1 local 192.168.4.0/24
set network tunnel ipsec IPSEC1 auto-key proxy-id ID1 remote
192.168.0.0/24
set network tunnel ipsec IPSEC1 tunnel-monitor enable no
set network tunnel ipsec IPSEC1 tunnel-interface tunnel.1
```

2.6.2 Web GUI configuration screenshot



The IPsec crypto profile setting is mapped to the default profile that the system automatically created (Step 4)



The Local Proxy ID and Remote Proxy ID are mapped to the local network and remote network respectively.

2.7 Setting the Virtual Router

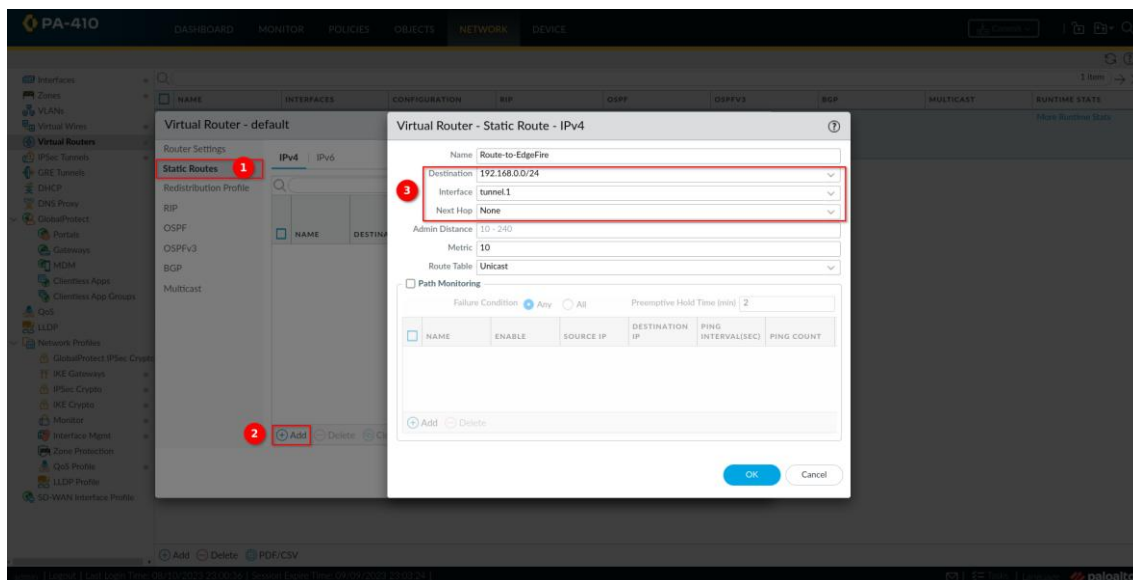
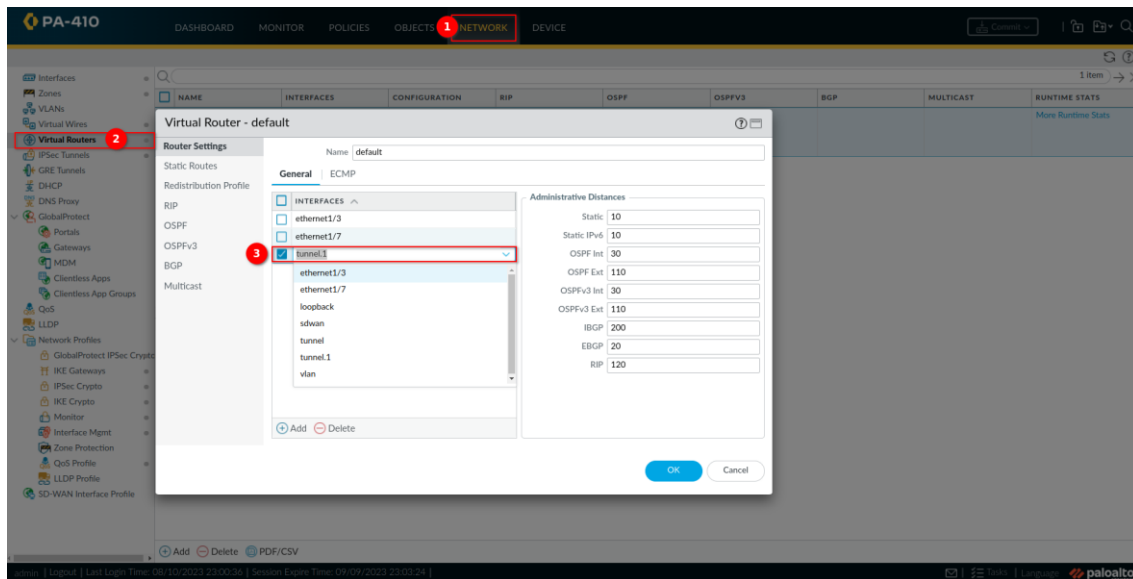
2.7.1 CLI Commands

```
set network virtual-router default protocol bgp enable no
set network virtual-router default protocol bgp dampening-profile default
cutoff 1.25
set network virtual-router default protocol bgp dampening-profile default
reuse 0.5
set network virtual-router default protocol bgp dampening-profile default
max-hold-time 900
set network virtual-router default protocol bgp dampening-profile default
decay-half-life-reachable 300
set network virtual-router default protocol bgp dampening-profile default
decay-half-life-unreachable 900
set network virtual-router default protocol bgp dampening-profile default
enable yes
set network virtual-router default protocol bgp routing-options graceful-
restart enable yes
set network virtual-router default protocol rip enable no
set network virtual-router default protocol ospf enable no
set network virtual-router default protocol ospfv3 enable no
set network virtual-router default interface [ ethernet1/3 ethernet1/7
tunnel.1 ]
set network virtual-router default ecmp algorithm ip-modulo

set network virtual-router default routing-table ip static-route Route-to-
EdgeFire interface tunnel.1
set network virtual-router default routing-table ip static-route Route-to-
EdgeFire metric 10
set network virtual-router default routing-table ip static-route Route-to-
EdgeFire destination 192.168.0.0/24
set network virtual-router default routing-table ip static-route Route-to-
EdgeFire route-table unicast

set network virtual-router default routing-table ip static-route Route-to-
EdgeFire path-monitor enable no
set network virtual-router default routing-table ip static-route Route-to-
EdgeFire path-monitor failure-condition any
set network virtual-router default routing-table ip static-route Route-to-
EdgeFire path-monitor hold-time 2
```

2.7.2 Web GUI configuration screenshot

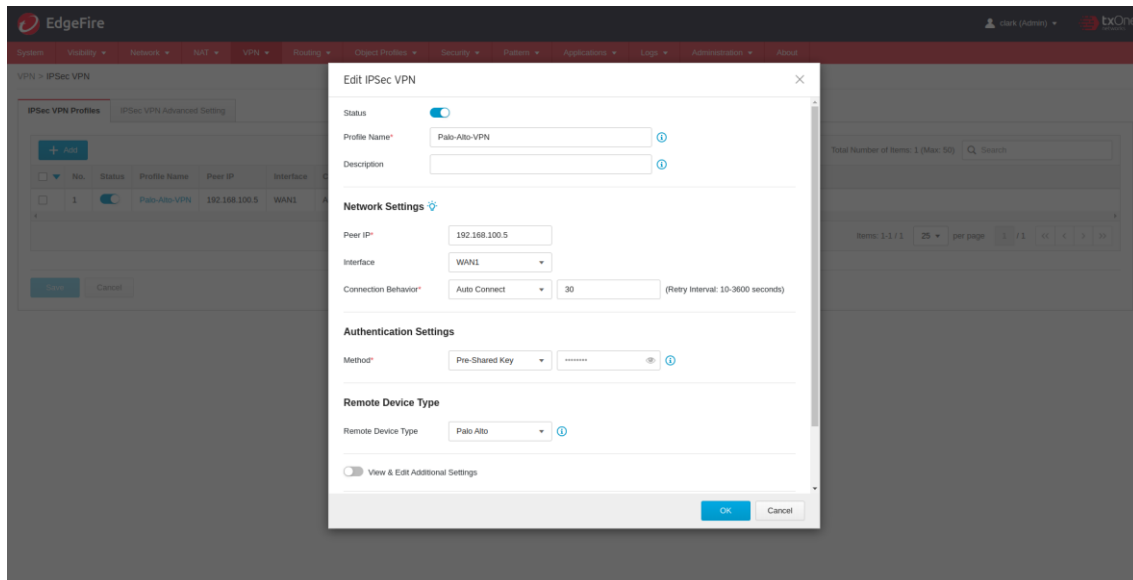


Add the tunnel interface to the virtual router and configure a static route to the remote VPN site

3 Setup flow for TXOne EdgeFire

3.1 IKE v1

1. Access web console -> VPN -> IPSec VPN -> IPSec VPN Profiles -> Create IPSec VPN



Pre-sharded key in this example is 1qaz2WSX

Remote Device Type

Remote Device Type Palo Alto i

☒ View & Edit Additional Settings

IKE Version 1 Peer Option

Negotiation Main Mode

Phase1

Proposal 1	Encryption	AES128	Authentication / PRF	SHA1	Diffie-Hellman Groups	2	—
Proposal 2	Encryption	AES128	Authentication / PRF	SHA256	Diffie-Hellman Groups	19	—
Proposal 3	Encryption	AES256	Authentication / PRF	SHA384	Diffie-Hellman Groups	20	—

Key Lifetime* 28800 120-172800 Seconds

Phase2

Local Interface* LAN1 (192.168.0.0/24) +

Remote IP/Subnet* IP Subnet 192.168.4.0 / 24 +

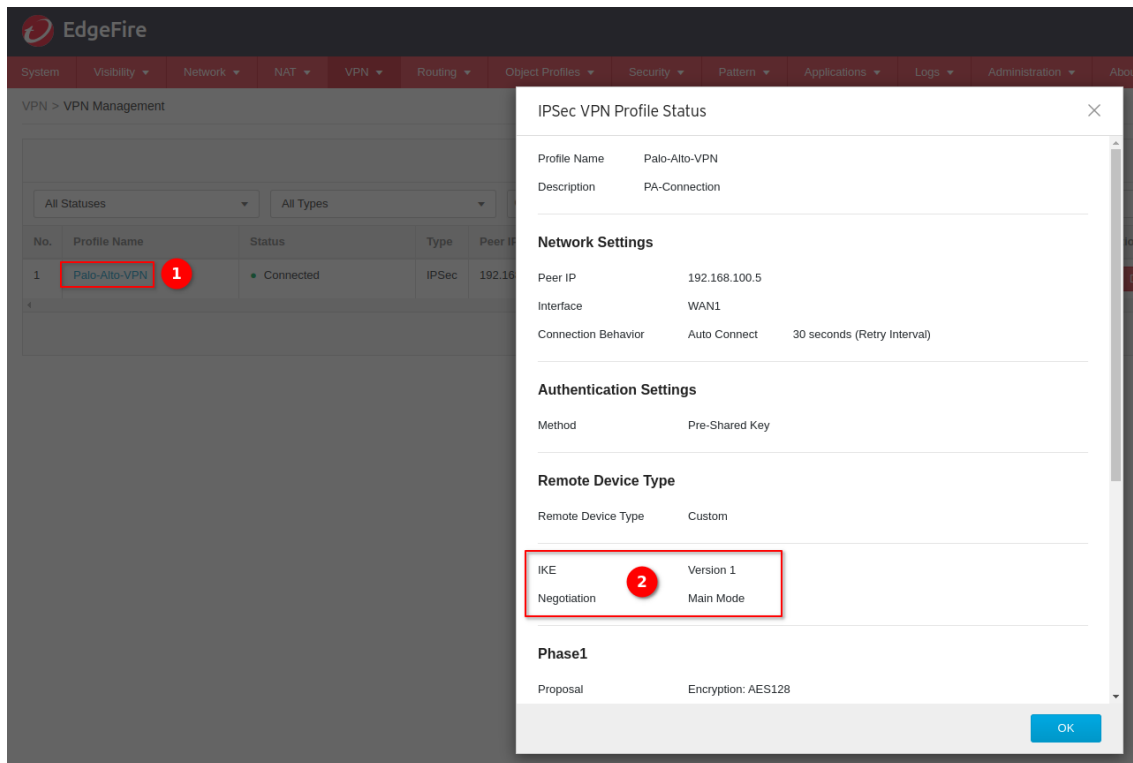
Proposal 1	Encryption	AES128	Authentication	SHA1	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 2	—
Proposal 2	Encryption	AES128G...	Authentication	Null	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 19	—
Proposal 3	Encryption	AES128G...	Authentication	Null	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 20	—

Key Lifetime* 3600 120-172800 Seconds

TXOne EdgeFire has a IPSec config template for default setting of Palo Alto crypto profile. You can click the “View & Edit Additional Settings” to review the details configuration of IKE Phase 1 and Phase 2.

The IKE settings should be mapped to the settings configured in Steps 3 and 4 for the Palo Alto device.

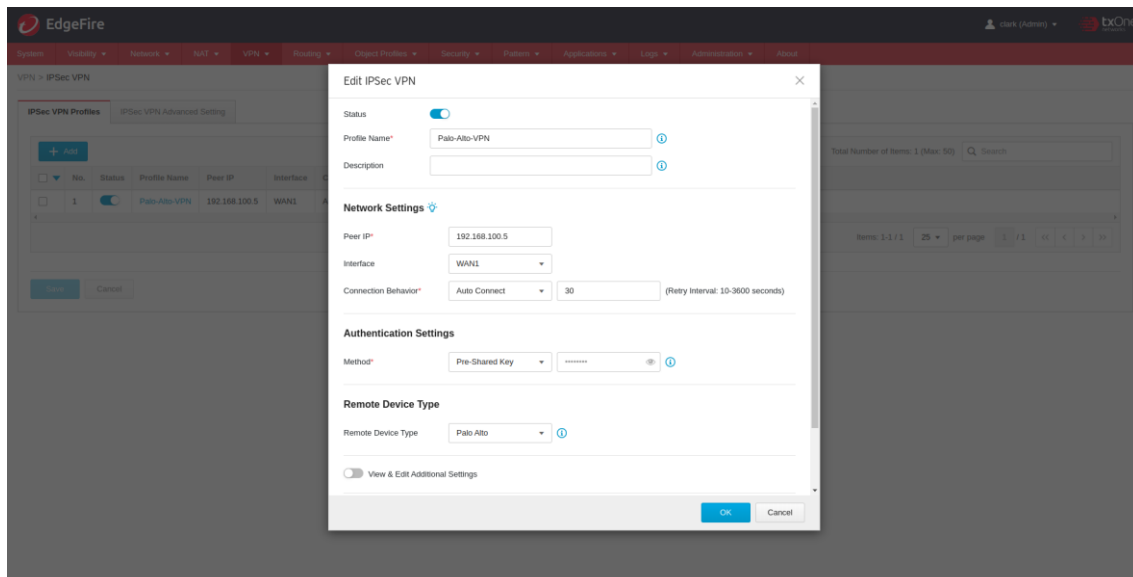
1. Access web console -> VPN -> VPN Management



You can check the VPN connection status in the VPN management page

3.2 IKE v2

1. Access web console -> VPN -> IPSec VPN -> IPSec VPN Profiles -> Create IPSec VPN



Pre-sharded key in this example is 1qaz2WSX

Remote Device Type

Remote Device Type Palo Alto i

☒ View & Edit Additional Settings

IKE Version 1 Peer Option

Negotiation Main Mode

Phase1

Proposal 1	Encryption	AES128	Authentication / PRF	SHA1	Diffie-Hellman Groups	2	—
Proposal 2	Encryption	AES128	Authentication / PRF	SHA256	Diffie-Hellman Groups	19	—
Proposal 3	Encryption	AES256	Authentication / PRF	SHA384	Diffie-Hellman Groups	20	—

Key Lifetime* 28800 120-172800 Seconds

Phase2

Local Interface* LAN1 (192.168.0.0/24) +

Remote IP/Subnet* IP Subnet 192.168.4.0 / 24 +

Proposal 1	Encryption	AES128	Authentication	SHA1	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 2	—
Proposal 2	Encryption	AES128G...	Authentication	Null	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 19	—
Proposal 3	Encryption	AES128G...	Authentication	Null	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 20	—

Key Lifetime* 3600 120-172800 Seconds

TXOne EdgeFire has a IPSec config template for default setting of Palo Alto crypto profile. You can click the “View & Edit Additional Settings” to review the details configuration of IKE Phase 1 and Phase 2. Then select the IKE version 2 in dropdown box.

☒ View & Edit Additional Settings

IKE
 Version 2
Peer Option

Phase1

Proposal 1	Encryption	AES128	Authentication / PRF	SHA1	Diffie-Hellman Groups	2	—
Proposal 2	Encryption	AES128	Authentication / PRF	SHA256	Diffie-Hellman Groups	19	—
Proposal 3	Encryption	AES256	Authentication / PRF	SHA384	Diffie-Hellman Groups	20	—

Key Lifetime* 28800 120-172800 Seconds

Phase2

Local Interface* LAN1 (192.168.0.0/24) +

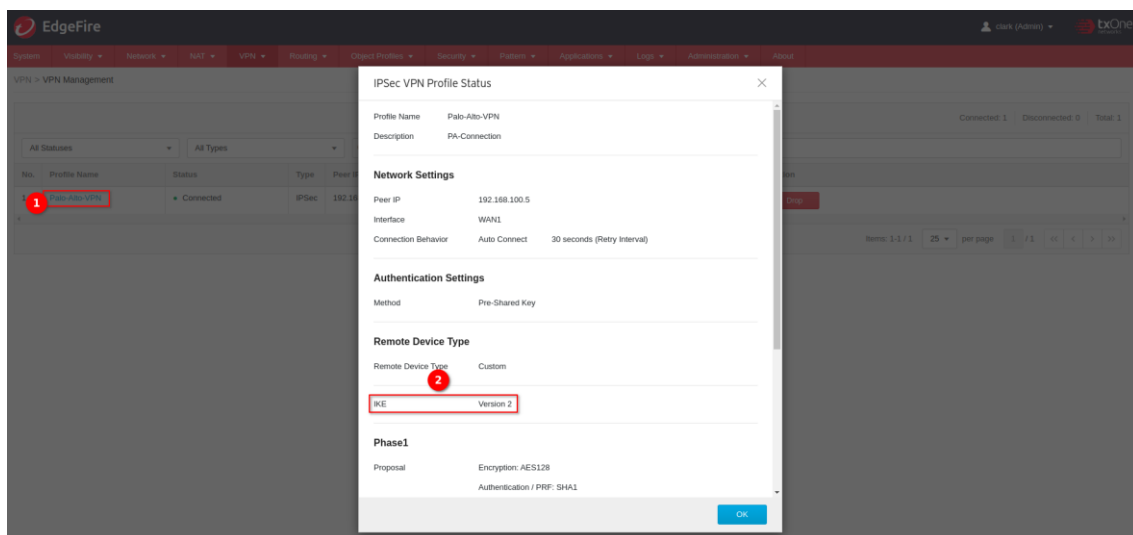
Remote IP/Subnet* IP Subnet 192.168.4.0 / 24 +

Proposal 1	Encryption	AES128	Authentication	SHA1	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 2	—
Proposal 2	Encryption	AES128G...	Authentication	Null	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 19	—
Proposal 3	Encryption	AES128G...	Authentication	Null	Perfect Forward Secrecy (PFS)	Diffie-Hellman Group 20	—

Key Lifetime* 3600 120-172800 Seconds

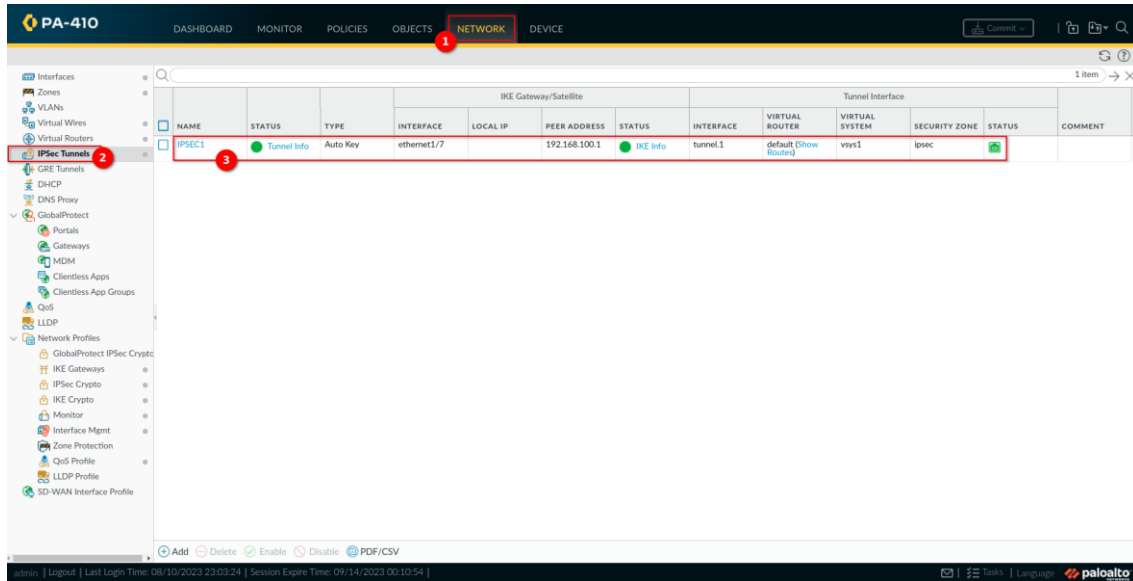
The IKE settings should be mapped to the settings configured in Steps 3 and 4 for the Palo Alto device.

1. Access web console -> VPN -> VPN Management



You can check the VPN connection status in the VPN management page

Screenshot for VPN status on Palo Alto site:



The screenshot shows the Palo Alto Networks PA-410 configuration interface. The 'NETWORK' tab is selected, and the 'IPsec Tunnels' section is highlighted in the left sidebar. A table displays the configuration for 'IPSEC1'.

			IKE Gateway/Satellite				Tunnel Interface					
NAME	STATUS	TYPE	INTERFACE	LOCAL IP	PEER ADDRESS	STATUS	INTERFACE	VIRTUAL ROUTER	VIRTUAL SYSTEM	SECURITY ZONE	STATUS	COMMENT
IPSEC1		Tunnel Info	Auto Key	ethernet1/7	192.168.100.1		tunnel.1	default [Show Routes]	vsys1	ipsec		

Annotations in the image:

- 1: Points to the 'NETWORK' tab in the top navigation bar.
- 2: Points to the 'IPsec Tunnels' section in the left sidebar.
- 3: Points to the 'IPSEC1' entry in the table.

At the bottom of the interface, there are links for 'Add', 'Delete', 'Enable', 'Disable', and 'PDF/CSV'. The footer shows the user 'sotras', login/logout options, last login time (08/10/2023 23:03:24), session expire time (09/14/2023 00:10:54), and the Palo Alto Networks logo.